Network Intrusion Detection Primer: Types of intrusion detection.

An intrusion detection system (IDS) monitors activity and raises alerts about malicious behaviour. A network-based IDS (NIDS) inspects traffic on a network segment, while a host-based IDS (HIDS) watches logs and system calls on one machine.

Signature-based detection matches traffic against patterns of known attacks. It is precise on known threats but blind to new ones. Anomaly-based detection learns what normal behaviour looks like and flags deviations, so it can find novel attacks but tends to produce more false positives.

Network Intrusion Detection Primer: The NSL-KDD dataset.

NSL-KDD is an improved version of the KDD Cup 99 intrusion dataset. It removes redundant records from the training set and duplicates from the test set so that classifiers are not biased toward frequent records. The training file KDDTrain+ contains 125,973 records and the test file KDDTest+ contains 22,544 records.

Each connection record has 41 features plus a label. Attacks fall into four families: denial of service (DoS), probing, remote to local (R2L) and user to root (U2R). The test set contains attack types that do not appear in the training set, which makes it a realistic test of generalisation.

Network Intrusion Detection Primer: Evaluating detectors.

Accuracy can be misleading for intrusion detection because of class imbalance: a classifier that labels every record as normal can still score a high accuracy when attacks are rare. Precision, recall and F1 are more informative.

The false positive rate matters in practice, since too many false alarms cause alert fatigue and real attacks get ignored. U2R is the rarest family and the hardest to detect, and a model that looks excellent on validation data can lose a lot of performance on the separate test set because of distribution shift.